

JULIEN GLORY MANANA

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PROFILE

Backend engineer building and shipping systems across African fintech and edtech. Designed multi-layer retrieval pipelines and LLM-integrated APIs; shipped deployed products with active users in education and healthtech. Focused on FastAPI, PostgreSQL, and LLM integration. Mastercard Foundation Scholar at KNUST.

SKILLS

Languages: Python (primary), Java, C++, SQL

Backend: FastAPI, Django, DRF, Flask, SQLAlchemy, PostgreSQL, REST APIs, Pydantic, Async

AI/ML: OpenAI, Claude, Deepseek, Gemini, RAG, Vector DBs (Pinecone)

DevOps/Tools: VS Code, IntelliJ, CLion, Git, GitHub, MS SQL Server, Poetry, Vercel, Render, Cursor, Claude Code

PROJECTS

1. 10-Day AI/ML & Agentic Systems Sprint — nGOT Giants · KNUST, Apr–May 2026 | [GitHub](#)

12 hrs/day · 8 projects across fintech, healthtech, agritech, and ecommerce

- **ETA Predictor** (Logistics) — ML model containerized with Docker, served via FastAPI; full experiment tracking via MLflow.
Stack: scikit-learn, FastAPI, Docker, MLflow, Pydantic V2, Poetry, pytest
- **Medical Literature RAG** (Healthtech) — Answers clinical questions with source citations; multi-step query decomposition over medical PDFs.
Stack: LlamaIndex, OpenAI GPT-4o-mini, text-embedding-3-small, Pinecone, FastAPI, Pydantic V2
- **Semantic Product Recommender** (eCommerce) — Natural-language queries ("cozy to wear in the rain") resolved via cosine similarity; returns top-N results with scores.
Stack: OpenAI text-embedding-3-small, Pinecone, FastAPI, Pydantic V2, Poetry
- **Telecom Complaint Analysis API** (Fintech/Telecom) — LLM function-calling pipeline with structured output and Pydantic validation; classifies and routes customer complaints end-to-end.
Stack: FastAPI, OpenAI function calling, Pydantic V2, Poetry
- **ReAct Agentic Systems** — Two LangChain ReAct agents with live tool integrations: a Satellite Intelligence API combining orbital calculations, mock data stores, and Tavily web search; and an eCommerce Shopper agent with real-time product ranking and structured tool-calling.
Stack: LangChain, Tavily Search, FastAPI, Pydantic V2, Poetry

2. Ocasia — Personalized Learning Platform | [GitHub](#) | [Live Demo](#)

Stack: Django, FastAPI, React, OpenAI GPT-4o-mini, Claude, PostgreSQL, RAG, Render, Vercel

- Generates personalized quizzes, flashcards, and AI explanations from student notes. Django as API gateway; async FastAPI service for agentic tasks and low-latency LLM calls.

3. VeriMed — Medicine Authenticity Risk Assessment (Experimental) | [GitHub](#) | [Live Demo](#)

Stack: FastAPI, Next.js, EasyOCR, pyzbar, rapidfuzz, OpenCV, PostgreSQL

- Three-image verification flow (packaging + barcode/QR) with OCR, fuzzy matching, and weighted risk scoring. AI follow-up assistant with session history persisted in Postgres.

4. ScholarAid — Scholarship Management & Application Platform | [GitHub](#) | [Live Demo](#)

Stack: Django, DRF, Claude API, JWT, PostgreSQL, BeautifulSoup4

- Claude extracts scholarship fields from URLs or raw text; scraper pipeline bulk-ingests portals. Students receive AI essay feedback powered by Claude.

5. Tool-Calling Chatbot — LLM Function-Calling DB Query | [GitHub](#)

Stack: FastAPI, SQLAlchemy, pyodbc, SQL Server, SQLite, basic RAG

- LLM decides when to call 6 predefined SQL Server functions vs RAG, and answers directly.

EXPERIENCE

Fellow

Feb 2026 – Present

Next Generation of Tech Giants (nGOT Giants) · Kumasi, Ghana

- Participated in a 10-day AI/ML and Agentic Systems intensive (May 2026) at KNUST's College of Engineering Innovation Centre — 12 hrs/day covering MLOps, RAG, fine-tuning, multi-agent systems, and production deployment.
- Building backend for Port, an internal collaboration tool, within a cross-functional engineering team.

Intern

Nov 2025 – Dec 2025

MTN Mobile Money Congo B. · Brazzaville, Republic of Congo

- Built a customer support chatbot prototype covering 15 MoMo service topics with a three-layer retrieval system: semantic search → keyword fallback → full context injection, eliminating hallucinations.
- Async FastAPI endpoints with semaphore-based concurrency; collaborated across product, BI, and ops teams inside a large-scale fintech operation.
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EDUCATION

BSc Computer Science

Expected Sept 2028

Kwame Nkrumah University of Science and Technology (KNUST) · Kumasi, Ghana

- Mastercard Foundation Scholar
- Relevant coursework: OOP (Java), C++, SQL, Operating Systems, Database Systems, System Analysis & Design, Numerical Methods, Probability & Statistics

W.A.S.S.C.E (Sciences)

Sep 2020 – Jun 2023

Rochas Foundation College of Africa · Nigeria

PROFESSIONAL TRAINING / CERTIFICATIONS

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| 1. 10-Day AI/ML & Agentic Systems Sprint | — nGOT Giants |
| 2. Associate AI Engineer for Developers | — Data Camp |
| 3. LangChain for LLM Applications | — DeepLearning.AI |
| 4. ChatGPT Prompt Engineering for Developers | — DeepLearning.AI |
| 5. Introduction to FastAPI | — Data Camp |
| 6. Python Backend Development with Django | — FreeCode Camp |

LEADERSHIP & COMMUNITY

Deputy Head of Technology

Sep 2025 – Jun 2026

Enactus KNUST

International Relations Officer

Jan 2026 – Sep 2026

Science Students' Association (SCISA) — KNUST

VALUENETWORK — Community sharing opportunities and resources for African students and builders.

FoundersLens — TikTok page (40K+ followers) sharing insights from top founders and builders.